DBMS MINIPROJECT

Peer-to-Peer Equipment Rental & Escrow-Based Booking System

Team number 13 :

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SQL Implementation :

- **Creating database**

```
mysql> CREATE DATABASE IF NOT EXISTS equipment_rental_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE equipment_rental_db;
Database changed
mysql>
```

- **CREATING TABLES**

```
mysql> CREATE TABLE Users (
    ->     user_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     name VARCHAR(100) NOT NULL,
    ->     email VARCHAR(100) UNIQUE NOT NULL,
    ->     phone VARCHAR(15),
    ->     password_hash VARCHAR(255) NOT NULL,
    ->     role ENUM('owner','renter','both') NOT NULL,
    ->     created_at DATETIME DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql>
mysql> DESC Users;
+---------------+-------------------------------+------+-----+-------------------+-------------------+
| Field         | Type                          | Null | Key | Default           | Extra             |
+---------------+-------------------------------+------+-----+-------------------+-------------------+
| user_id       | int                           | NO   | PRI | NULL              | auto_increment    |
| name          | varchar(100)                  | NO   |     | NULL              |                   |
| email         | varchar(100)                  | NO   | UNI | NULL              |                   |
| phone         | varchar(15)                   | YES  |     | NULL              |                   |
| password_hash | varchar(255)                  | NO   |     | NULL              |                   |
| role          | enum('owner','renter','both') | NO   |     | NULL              |                   |
| created_at    | datetime                      | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+---------------+-------------------------------+------+-----+-------------------+-------------------+
7 rows in set (0.01 sec)
```

```
mysql> CREATE TABLE Owner_Profile (
    ->     owner_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     user_id INT NOT NULL,
    ->     bank_account_no VARCHAR(30),
    ->     ifsc_code VARCHAR(15),
    ->     verified BOOLEAN DEFAULT FALSE,
    ->     CONSTRAINT fk_owner_user FOREIGN KEY (user_id)
    ->         REFERENCES Users(user_id) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DESC Owner_Profile;
+-----------------+-------------+------+-----+---------+----------------+
| Field           | Type        | Null | Key | Default | Extra          |
+-----------------+-------------+------+-----+---------+----------------+
| owner_id        | int         | NO   | PRI | NULL    | auto_increment |
| user_id         | int         | NO   | MUL | NULL    |                |
| bank_account_no | varchar(30) | YES  |     | NULL    |                |
| ifsc_code       | varchar(15) | YES  |     | NULL    |                |
| verified        | tinyint(1)  | YES  |     | 0       |                |
+-----------------+-------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Renter_Profile (
    ->     renter_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     user_id INT NOT NULL,
    ->     id_proof_type VARCHAR(30),
    ->     id_proof_number VARCHAR(50),
    ->     verified BOOLEAN DEFAULT FALSE,
    ->     CONSTRAINT fk_renter_user FOREIGN KEY (user_id)
    ->         REFERENCES Users(user_id) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DESC Renter_Profile;
+-----------------+-------------+------+-----+---------+----------------+
| Field           | Type        | Null | Key | Default | Extra          |
+-----------------+-------------+------+-----+---------+----------------+
| renter_id       | int         | NO   | PRI | NULL    | auto_increment |
| user_id         | int         | NO   | MUL | NULL    |                |
| id_proof_type   | varchar(30) | YES  |     | NULL    |                |
| id_proof_number | varchar(50) | YES  |     | NULL    |                |
| verified        | tinyint(1)  | YES  |     | 0       |                |
+-----------------+-------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Equipment_Category (
    ->     category_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     category_name VARCHAR(50) NOT NULL,
    ->     description VARCHAR(255)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DESC Equipment_Category;
+---------------+--------------+------+-----+---------+----------------+
| Field         | Type         | Null | Key | Default | Extra          |
+---------------+--------------+------+-----+---------+----------------+
| category_id   | int          | NO   | PRI | NULL    | auto_increment |
| category_name | varchar(50)  | NO   |     | NULL    |                |
| description   | varchar(255) | YES  |     | NULL    |                |
+---------------+--------------+------+-----+---------+----------------+
3 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Equipment (
    ->     equipment_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     owner_id INT NOT NULL,
    ->     category_id INT NOT NULL,
    ->     title VARCHAR(100) NOT NULL,
    ->     description TEXT,
    ->     price_per_day DECIMAL(10,2) NOT NULL CHECK (price_per_day > 0),
    ->     condition_status ENUM('new','good','fair') DEFAULT 'good',
    ->     availability_status ENUM('available','booked','maintenance') DEFAULT 'available',
    ->     created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     CONSTRAINT fk_equipment_owner FOREIGN KEY (owner_id)
    ->         REFERENCES Owner_Profile(owner_id) ON DELETE CASCADE,
    ->     CONSTRAINT fk_equipment_category FOREIGN KEY (category_id)
    ->         REFERENCES Equipment_Category(category_id)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DESC Equipment;
+--------------------+-----------------------------------------+------+-----+-------------------+-------------------+
| Field              | Type                                    | Null | Key | Default           | Extra             |
+--------------------+-----------------------------------------+------+-----+-------------------+-------------------+
| equipment_id       | int                                     | NO   | PRI | NULL              | auto_increment    |
| owner_id           | int                                     | NO   | MUL | NULL              |                   |
| category_id        | int                                     | NO   | MUL | NULL              |                   |
| title              | varchar(100)                            | NO   |     | NULL              |                   |
| description        | text                                    | YES  |     | NULL              |                   |
| price_per_day      | decimal(10,2)                           | NO   |     | NULL              |                   |
| condition_status   | enum('new','good','fair')               | YES  |     | good              |                   |
| availability_status | enum('available','booked','maintenance') | YES  |     | available         |                   |
| created_at         | datetime                                | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+--------------------+-----------------------------------------+------+-----+-------------------+-------------------+
9 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Booking (
    ->     booking_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     equipment_id INT NOT NULL,
    ->     renter_id INT NOT NULL,
    ->     start_date DATE NOT NULL,
    ->     end_date DATE NOT NULL,
    ->     total_amount DECIMAL(10,2),
    ->     status ENUM('pending','confirmed','ongoing','returned','cancelled') DEFAULT 'pending',
    ->     created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     CONSTRAINT fk_booking_equipment FOREIGN KEY (equipment_id)
    ->         REFERENCES Equipment(equipment_id),
    ->     CONSTRAINT fk_booking_renter FOREIGN KEY (renter_id)
    ->         REFERENCES Renter_Profile(renter_id),
    ->     CONSTRAINT chk_booking_dates CHECK (end_date >= start_date)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DESC Booking;
+-------------+-----------------------------------------------------------------+------+-----+-------------------+-------------------+
| Field       | Type                                                            | Null | Key | Default           | Extra             |
+-------------+-----------------------------------------------------------------+------+-----+-------------------+-------------------+
| booking_id  | int                                                             | NO   | PRI | NULL              | auto_increment    |
| equipment_id | int                                                            | NO   | MUL | NULL              |                   |
| renter_id   | int                                                             | NO   | MUL | NULL              |                   |
| start_date  | date                                                            | NO   |     | NULL              |                   |
| end_date    | date                                                            | NO   |     | NULL              |                   |
| total_amount | decimal(10,2)                                                  | YES  |     | NULL              |                   |
| status      | enum('pending','confirmed','ongoing','returned','cancelled')    | YES  |     | pending           |                   |
| created_at  | datetime                                                        | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-------------+-----------------------------------------------------------------+------+-----+-------------------+-------------------+
8 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Booking_Status_Log (
    ->     log_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     booking_id INT NOT NULL,
    ->     old_status VARCHAR(20),
    ->     new_status VARCHAR(20) NOT NULL,
    ->     changed_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     CONSTRAINT fk_log_booking FOREIGN KEY (booking_id)
    ->         REFERENCES Booking(booking_id) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DESC Booking_Status_Log;
+-------------+-------------+------+-----+-------------------+-------------------+
| Field       | Type        | Null | Key | Default           | Extra             |
+-------------+-------------+------+-----+-------------------+-------------------+
| log_id      | int         | NO   | PRI | NULL              | auto_increment    |
| booking_id  | int         | NO   | MUL | NULL              |                   |
| old_status  | varchar(20) | YES  |     | NULL              |                   |
| new_status  | varchar(20) | NO   |     | NULL              |                   |
| changed_at  | datetime    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-------------+-------------+------+-----+-------------------+-------------------+
5 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Payment (
    ->     payment_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     booking_id INT NOT NULL UNIQUE,
    ->     amount DECIMAL(10,2) NOT NULL,
    ->     payment_method ENUM('card','upi','netbanking') NOT NULL,
    ->     payment_date DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     CONSTRAINT fk_payment_booking FOREIGN KEY (booking_id)
    ->         REFERENCES Booking(booking_id) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DESC Payment;
+----------------+---------------------------------+------+-----+-------------------+-------------------+
| Field          | Type                            | Null | Key | Default           | Extra             |
+----------------+---------------------------------+------+-----+-------------------+-------------------+
| payment_id     | int                             | NO   | PRI | NULL              | auto_increment    |
| booking_id     | int                             | NO   | UNI | NULL              |                   |
| amount         | decimal(10,2)                   | NO   |     | NULL              |                   |
| payment_method | enum('card','upi','netbanking') | NO   |     | NULL              |                   |
| payment_date   | datetime                        | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+----------------+---------------------------------+------+-----+-------------------+-------------------+
5 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Escrow_Transaction (
    ->     escrow_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     payment_id INT NOT NULL UNIQUE,
    ->     status ENUM('held','released','refunded','disputed') DEFAULT 'held',
    ->     held_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     released_at DATETIME NULL,
    ->     CONSTRAINT fk_escrow_payment FOREIGN KEY (payment_id)
    ->         REFERENCES Payment(payment_id) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DESC Escrow_Transaction;
+-------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| Field       | Type                                          | Null | Key | Default           | Extra             |
+-------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| escrow_id   | int                                           | NO   | PRI | NULL              | auto_increment    |
| payment_id  | int                                           | NO   | UNI | NULL              |                   |
| status      | enum('held','released','refunded','disputed') | YES  |     | held              |                   |
| held_at     | datetime                                      | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| released_at | datetime                                      | YES  |     | NULL              |                   |
+-------------+-----------------------------------------------+------+-----+-------------------+-------------------+
5 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Dispute (
    ->     dispute_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     booking_id INT NOT NULL,
    ->     raised_by INT NOT NULL,
    ->     reason VARCHAR(255) NOT NULL,
    ->     status ENUM('open','under_review','resolved') DEFAULT 'open',
    ->     created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     CONSTRAINT fk_dispute_booking FOREIGN KEY (booking_id)
    ->         REFERENCES Booking(booking_id) ON DELETE CASCADE,
    ->     CONSTRAINT fk_dispute_user FOREIGN KEY (raised_by)
    ->         REFERENCES Users(user_id)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DESC Dispute;
+------------+---------------------------------------+------+-----+-------------------+-------------------+
| Field      | Type                                  | Null | Key | Default           | Extra             |
+------------+---------------------------------------+------+-----+-------------------+-------------------+
| dispute_id | int                                   | NO   | PRI | NULL              | auto_increment    |
| booking_id | int                                   | NO   | MUL | NULL              |                   |
| raised_by  | int                                   | NO   | MUL | NULL              |                   |
| reason     | varchar(255)                          | NO   |     | NULL              |                   |
| status     | enum('open','under_review','resolved') | YES  |     | open              |                   |
| created_at | datetime                              | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------+---------------------------------------+------+-----+-------------------+-------------------+
6 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Review (
    ->     review_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     booking_id INT NOT NULL,
    ->     reviewer_id INT NOT NULL,
    ->     rating INT NOT NULL CHECK (rating BETWEEN 1 AND 5),
    ->     comment VARCHAR(255),
    ->     created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     CONSTRAINT fk_review_booking FOREIGN KEY (booking_id)
    ->         REFERENCES Booking(booking_id) ON DELETE CASCADE,
    ->     CONSTRAINT fk_review_user FOREIGN KEY (reviewer_id)
    ->         REFERENCES Users(user_id)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DESC Review;
+-------------+--------------+------+-----+-------------------+-------------------+
| Field       | Type         | Null | Key | Default           | Extra             |
+-------------+--------------+------+-----+-------------------+-------------------+
| review_id   | int          | NO   | PRI | NULL              | auto_increment    |
| booking_id  | int          | NO   | MUL | NULL              |                   |
| reviewer_id | int          | NO   | MUL | NULL              |                   |
| rating      | int          | NO   |     | NULL              |                   |
| comment     | varchar(255) | YES  |     | NULL              |                   |
| created_at  | datetime     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-------------+--------------+------+-----+-------------------+-------------------+
6 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Damage_Report (
    ->     report_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     booking_id INT NOT NULL,
    ->     description TEXT NOT NULL,
    ->     estimated_cost DECIMAL(10,2),
    ->     reported_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    ->     CONSTRAINT fk_damage_booking FOREIGN KEY (booking_id)
    ->         REFERENCES Booking(booking_id) ON DELETE CASCADE
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DESC Damage_Report;
+----------------+---------------+------+-----+-------------------+-------------------+
| Field          | Type          | Null | Key | Default           | Extra             |
+----------------+---------------+------+-----+-------------------+-------------------+
| report_id      | int           | NO   | PRI | NULL              | auto_increment    |
| booking_id     | int           | NO   | MUL | NULL              |                   |
| description    | text          | NO   |     | NULL              |                   |
| estimated_cost | decimal(10,2) | YES  |     | NULL              |                   |
| reported_at    | datetime      | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+----------------+---------------+------+-----+-------------------+-------------------+
5 rows in set (0.00 sec)
```

- **ALTERING THE TABLES**

```
mysql> ALTER TABLE Escrow_Transaction
    -> ADD COLUMN platform_fee_percent DECIMAL(5,2) NOT NULL DEFAULT 5.00;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> ALTER TABLE Escrow_Transaction
    -> ADD COLUMN platform_fee_amount DECIMAL(10,2) DEFAULT 0.00;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

- **VIEWS**

```
mysql> CREATE OR REPLACE VIEW Available_Equipment_View AS
    -> SELECT
    ->     e.equipment_id,
    ->     e.title,
    ->     e.description,
    ->     ec.category_name,
    ->     u.name AS owner_name,
    ->     e.price_per_day,
    ->     e.condition_status,
    ->     e.availability_status
    -> FROM Equipment e
    -> JOIN Owner_Profile op ON e.owner_id = op.owner_id
    -> JOIN Users u ON op.user_id = u.user_id
    -> JOIN Equipment_Category ec ON e.category_id = ec.category_id
    -> WHERE e.availability_status = 'available';
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE OR REPLACE VIEW Booking_Escrow_View AS
    -> SELECT
    ->     b.booking_id,
    ->     e.title AS equipment,
    ->     ru.name AS renter_name,
    ->     b.start_date,
    ->     b.end_date,
    ->     b.total_amount,
    ->     b.status AS booking_status,
    ->     p.payment_id,
    ->     p.amount AS paid_amount,
    ->     et.escrow_id,
    ->     et.status AS escrow_status,
    ->     et.platform_fee_percent,
    ->     et.platform_fee_amount
    -> FROM Booking b
    -> JOIN Equipment e ON b.equipment_id = e.equipment_id
    -> JOIN Renter_Profile rp ON b.renter_id = rp.renter_id
    -> JOIN Users ru ON rp.user_id = ru.user_id
    -> LEFT JOIN Payment p ON b.booking_id = p.booking_id
    -> LEFT JOIN Escrow_Transaction et ON p.payment_id = et.payment_id;
Query OK, 0 rows affected (0.00 sec)
```

- **INSERTING VALUES**

```
mysql> INSERT INTO Users (name, email, phone, password_hash, role)
    -> VALUES
    -> ('Arjun Kumar', 'arjun@example.com', '9876543210',
    ->  '$2b$12$ExampleHashForArjun12345678901234567890123456789', 'owner'),
    -> ('Meera Sharma', 'meera@example.com', '9876543211',
    ->  '$2b$12$ExampleHashForMeera12345678901234567890123456789', 'renter'),
    -> ('Rahul Verma', 'rahul@example.com', '9876543212',
    ->  '$2b$12$ExampleHashForRahul12345678901234567890123456789', 'both'),
    -> ('Ananya Rao', 'ananya@example.com', '9876543213',
    ->  '$2b$12$ExampleHashForAnanya12345678901234567890123456789', 'both');
Query OK, 4 rows affected (0.00 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Owner_Profile (user_id, bank_account_no, ifsc_code, verified)
    -> SELECT user_id, 'XXXXXX1234', 'SBIN0001234', TRUE
    -> FROM Users WHERE email = 'arjun@example.com';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Owner_Profile (user_id, bank_account_no, ifsc_code, verified)
    -> SELECT user_id, 'XXXXXX5678', 'HDFC0005678', TRUE
    -> FROM Users WHERE email = 'rahul@example.com';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> INSERT INTO Owner_Profile (user_id, bank_account_no, ifsc_code, verified)
    -> SELECT user_id, 'XXXXXX9012', 'ICIC0009012', TRUE
    -> FROM Users WHERE email = 'ananya@example.com';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Renter_Profile (user_id, id_proof_type, id_proof_number, verified)
    -> SELECT user_id, 'Aadhaar', 'XXXX-XXXX-1001', TRUE
    -> FROM Users WHERE email = 'meera@example.com';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Renter_Profile (user_id, id_proof_type, id_proof_number, verified)
    -> SELECT user_id, 'Aadhaar', 'XXXX-XXXX-1002', TRUE
    -> FROM Users WHERE email = 'rahul@example.com';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Renter_Profile (user_id, id_proof_type, id_proof_number, verified)
    -> SELECT user_id, 'Driving License', 'DL-XXXX-1003', TRUE
    -> FROM Users WHERE email = 'ananya@example.com';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Equipment_Category (category_name, description)
    -> VALUES
    -> ('Photography & Videography', 'Cameras, drones, tripods and video equipment'),
    -> ('Power Tools & DIY', 'Power tools and equipment for repair and DIY work'),
    -> ('Camping & Outdoor', 'Camping and outdoor recreation equipment'),
    -> ('Sports & Fitness', 'Sports and fitness equipment'),
    -> ('Party & Events', 'Equipment used for parties and events'),
    -> ('Electronics', 'Laptops, gaming consoles, VR headsets and projectors');
Query OK, 6 rows affected (0.00 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Equipment
    -> (owner_id, category_id, title, description, price_per_day, condition_status, availability_status)
    -> SELECT op.owner_id, ec.category_id, 'Canon DSLR Camera',
    ->        'Professional DSLR camera suitable for photography and video recording.',
    ->        1200.00, 'good', 'available'
    -> FROM Owner_Profile op
    -> JOIN Users u ON op.user_id = u.user_id
    -> JOIN Equipment_Category ec ON ec.category_name = 'Photography & Videography'
    -> WHERE u.email = 'arjun@example.com';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Equipment
    -> (owner_id, category_id, title, description, price_per_day, condition_status, availability_status)
    -> SELECT op.owner_id, ec.category_id, 'Mountain Bike',
    ->        'Well-maintained mountain bike suitable for outdoor trips.',
    ->        800.00, 'good', 'available'
    -> FROM Owner_Profile op
    -> JOIN Users u ON op.user_id = u.user_id
    -> JOIN Equipment_Category ec ON ec.category_name = 'Sports & Fitness'
    -> WHERE u.email = 'rahul@example.com';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Equipment
    -> (owner_id, category_id, title, description, price_per_day, condition_status, availability_status)
    -> SELECT op.owner_id, ec.category_id, 'Gaming Console',
    ->        'Latest generation gaming console with two controllers.',
    ->        1000.00, 'new', 'available'
    -> FROM Owner_Profile op
    -> JOIN Users u ON op.user_id = u.user_id
    -> JOIN Equipment_Category ec ON ec.category_name = 'Electronics'
    -> WHERE u.email = 'ananya@example.com';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Equipment
    -> (owner_id, category_id, title, description, price_per_day, condition_status, availability_status)
    -> SELECT op.owner_id, ec.category_id, 'Camping Tent',
    ->        'Four-person waterproof camping tent.',
    ->        500.00, 'good', 'available'
    -> FROM Owner_Profile op
    -> JOIN Users u ON op.user_id = u.user_id
    -> JOIN Equipment_Category ec ON ec.category_name = 'Camping & Outdoor'
    -> WHERE u.email = 'arjun@example.com';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Booking
    -> (equipment_id, renter_id, start_date, end_date, total_amount, status)
    -> SELECT e.equipment_id, rp.renter_id,
    ->        '2026-10-05', '2026-10-07',
    ->        e.price_per_day * (DATEDIFF('2026-10-07', '2026-10-05') + 1),
    ->        'confirmed'
    -> FROM Equipment e
    -> JOIN Renter_Profile rp
    -> JOIN Users u ON rp.user_id = u.user_id
    -> WHERE e.title = 'Canon DSLR Camera'
    ->   AND u.email = 'meera@example.com';
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0
```

- **UPDATING THE TABLES**

```
mysql> UPDATE Equipment e
    -> JOIN Booking b ON e.equipment_id = b.equipment_id
    -> SET e.availability_status = 'booked'
    -> WHERE b.status IN ('confirmed', 'ongoing')
    ->   AND b.start_date <= '2026-10-07'
    ->   AND b.end_date >= '2026-10-05';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

- **INSERTION**

```
mysql> INSERT INTO Payment (booking_id, amount, payment_method)
    -> SELECT booking_id, total_amount, 'upi'
    -> FROM Booking
    -> WHERE equipment_id = (
    ->     SELECT equipment_id
    ->     FROM Equipment
    ->     WHERE title = 'Canon DSLR Camera'
    ->     LIMIT 1
    -> )
    -> AND renter_id = (
    ->     SELECT renter_id
    ->     FROM Renter_Profile rp
    ->     JOIN Users u ON rp.user_id = u.user_id
    ->     WHERE u.email = 'meera@example.com'
    ->     LIMIT 1
    -> )
    -> ORDER BY booking_id DESC
    -> LIMIT 1;
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Escrow_Transaction
    -> (payment_id, status, platform_fee_percent, platform_fee_amount)
    -> SELECT payment_id,
    ->        'held',
    ->        5.00,
    ->        ROUND(amount * 0.05, 2)
    -> FROM Payment
    -> ORDER BY payment_id DESC
    -> LIMIT 1;
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0
```

- **SELECT : basic retrieval**

```
mysql> SELECT equipment_id, title, price_per_day, condition_status
    -> FROM Equipment
    -> WHERE availability_status = 'available'
    -> ORDER BY price_per_day ASC;
+--------------+----------------+--------------+------------------+
| equipment_id | title          | price_per_day | condition_status |
+--------------+----------------+--------------+------------------+
|            4 | Camping Tent   |        500.00 | good             |
|            2 | Mountain Bike  |        800.00 | good             |
|            3 | Gaming Console |       1000.00 | new              |
+--------------+----------------+--------------+------------------+
3 rows in set (0.00 sec)
```

- **JOIN: equipment + owner + category**

```
mysql> SELECT
    ->     e.title,
    ->     ec.category_name,
    ->     u.name AS owner_name,
    ->     e.price_per_day,
    ->     e.condition_status
    -> FROM Equipment e
    -> JOIN Owner_Profile op ON e.owner_id = op.owner_id
    -> JOIN Users u ON op.user_id = u.user_id
    -> JOIN Equipment_Category ec ON e.category_id = ec.category_id
    -> ORDER BY e.price_per_day DESC;
+-----------------+--------------------------+-------------+--------------+------------------+
| title           | category_name            | owner_name  | price_per_day | condition_status |
+-----------------+--------------------------+-------------+--------------+------------------+
| Canon DSLR Camera | Photography & Videography | Arjun Kumar |       1200.00 | good             |
| Gaming Console  | Electronics              | Ananya Rao  |       1000.00 | new              |
| Mountain Bike   | Sports & Fitness         | Rahul Verma |        800.00 | good             |
| Camping Tent    | Camping & Outdoor        | Arjun Kumar |        500.00 | good             |
+-----------------+--------------------------+-------------+--------------+------------------+
4 rows in set (0.00 sec)
```

- **GROUP BY +AGGREGAE**
    - **Number of equipment items in each category**

```
mysql> SELECT
    ->     ec.category_name,
    ->     COUNT(e.equipment_id) AS equipment_count,
    ->     ROUND(AVG(e.price_per_day), 2) AS average_daily_price,
    ->     MAX(e.price_per_day) AS maximum_daily_price
    -> FROM Equipment_Category ec
    -> LEFT JOIN Equipment e ON ec.category_id = e.category_id
    -> GROUP BY ec.category_id, ec.category_name
    -> ORDER BY equipment_count DESC;
+--------------------------+-----------------+---------------------+---------------------+
| category_name            | equipment_count | average_daily_price | maximum_daily_price |
+--------------------------+-----------------+---------------------+---------------------+
| Photography & Videography |               1 |             1200.00 |             1200.00 |
| Camping & Outdoor         |               1 |              500.00 |              500.00 |
| Sports & Fitness          |               1 |              800.00 |              800.00 |
| Electronics               |               1 |             1000.00 |             1000.00 |
| Power Tools & DIY         |               0 |                NULL |                NULL |
| Party & Events            |               0 |                NULL |                NULL |
+--------------------------+-----------------+---------------------+---------------------+
6 rows in set (0.00 sec)
```

- **HAVING**
    - **Categories containing at least one equipment item**
    - **with average price above 700**

```
mysql> SELECT
    ->     ec.category_name,
    ->     ROUND(AVG(e.price_per_day), 2) AS average_price
    -> FROM Equipment e
    -> JOIN Equipment_Category ec ON e.category_id = ec.category_id
    -> GROUP BY ec.category_id, ec.category_name
    -> HAVING AVG(e.price_per_day) > 700
    -> ORDER BY average_price DESC;
+--------------------------+---------------+
| category_name            | average_price |
+--------------------------+---------------+
| Photography & Videography |       1200.00 |
| Electronics               |       1000.00 |
| Sports & Fitness          |        800.00 |
+--------------------------+---------------+
3 rows in set (0.00 sec)
```

- **Aggregate revenue**

```
mysql> SELECT
    ->     COUNT(*) AS total_payments,
    ->     SUM(amount) AS total_payment_amount,
    ->     AVG(amount) AS average_payment_amount,
    ->     MIN(amount) AS minimum_payment,
    ->     MAX(amount) AS maximum_payment
    -> FROM Payment;
+----------------+----------------------+-----------------------+-----------------+-----------------+
| total_payments | total_payment_amount | average_payment_amount | minimum_payment | maximum_payment |
+----------------+----------------------+-----------------------+-----------------+-----------------+
|              1 |              3600.00 |            3600.000000 |         3600.00 |         3600.00 |
+----------------+----------------------+-----------------------+-----------------+-----------------+
1 row in set (0.00 sec)
```

- **Multi-table JOIN**
    - **Booking, renter, equipment, payment and escrow**

```
mysql> SELECT
    ->     b.booking_id,
    ->     e.title AS equipment,
    ->     u.name AS renter,
    ->     b.start_date,
    ->     b.end_date,
    ->     b.total_amount,
    ->     b.status AS booking_status,
    ->     p.amount AS paid_amount,
    ->     et.status AS escrow_status
    -> FROM Booking b
    -> JOIN Equipment e ON b.equipment_id = e.equipment_id
    -> JOIN Renter_Profile rp ON b.renter_id = rp.renter_id
    -> JOIN Users u ON rp.user_id = u.user_id
    -> LEFT JOIN Payment p ON b.booking_id = p.booking_id
    -> LEFT JOIN Escrow_Transaction et ON p.payment_id = et.payment_id
    -> ORDER BY b.created_at DESC;
+------------+------------------+--------------+------------+------------+--------------+----------------+-------------+---------------+
| booking_id | equipment        | renter       | start_date | end_date   | total_amount | booking_status | paid_amount | escrow_status |
+------------+------------------+--------------+------------+------------+--------------+----------------+-------------+---------------+
|          1 | Canon DSLR Camera | Meera Sharma | 2026-10-05 | 2026-10-07 |      3600.00 | confirmed      |     3600.00 | held          |
+------------+------------------+--------------+------------+------------+--------------+----------------+-------------+---------------+
1 row in set (0.00 sec)
```

- **SUBQUERY**

```
mysql> SELECT equipment_id, title, price_per_day
    -> FROM Equipment
    -> WHERE price_per_day > (
    ->     SELECT AVG(price_per_day)
    ->     FROM Equipment
    -> )
    -> ORDER BY price_per_day DESC;
+--------------+--------------------+---------------+
| equipment_id | title              | price_per_day |
+--------------+--------------------+---------------+
|            1 | Canon DSLR Camera  |       1200.00 |
|            3 | Gaming Console     |       1000.00 |
+--------------+--------------------+---------------+
2 rows in set (0.00 sec)
```

- **CORRELATED SUBQUERY**

```
mysql> SELECT e.equipment_id, e.title, e.price_per_day, e.category_id
    -> FROM Equipment e
    -> WHERE e.price_per_day > (
    ->     SELECT AVG(e2.price_per_day)
    ->     FROM Equipment e2
    ->     WHERE e2.category_id = e.category_id
    -> )
    -> ORDER BY e.category_id, e.price_per_day DESC;
Empty set (0.00 sec)
```

- **EXISTS**
    - **Owners who currently have at least one listed equipment item**

```
mysql> SELECT u.user_id, u.name, u.email
    -> FROM Users u
    -> WHERE EXISTS (
    ->     SELECT 1
    ->     FROM Owner_Profile op
    ->     JOIN Equipment e ON op.owner_id = e.owner_id
    ->     WHERE op.user_id = u.user_id
    -> );
+---------+-------------+--------------------+
| user_id | name        | email              |
+---------+-------------+--------------------+
|       1 | Arjun Kumar | arjun@example.com  |
|       3 | Rahul Verma | rahul@example.com  |
|       4 | Ananya Rao  | ananya@example.com |
+---------+-------------+--------------------+
3 rows in set (0.00 sec)
```

- **DELETE demonstration**

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Equipment_Category (category_name, description)
    -> VALUES ('Temporary Demo Category', 'Used only to demonstrate DELETE');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> DELETE FROM Equipment_Category
    -> WHERE category_id = (
    ->     SELECT category_id
    ->     FROM (
    ->         SELECT category_id
    ->         FROM Equipment_Category
    ->         WHERE category_name = 'Temporary Demo Category'
    ->     ) AS temp_category
    -> );
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)
```

- **TRANSACTION**

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Booking
    -> (equipment_id, renter_id, start_date, end_date, total_amount, status)
    -> SELECT
    ->     e.equipment_id,
    ->     rp.renter_id,
    ->     '2026-10-10',
    ->     '2026-10-12',
    ->     e.price_per_day * (DATEDIFF('2026-10-12', '2026-10-10') + 1),
    ->     'confirmed'
    -> FROM Equipment e
    -> JOIN Renter_Profile rp
    -> JOIN Users u ON rp.user_id = u.user_id
    -> WHERE e.title = 'Mountain Bike'
    ->   AND u.email = 'meera@example.com'
    -> LIMIT 1;
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0

mysql>
mysql> SET @new_booking_id = LAST_INSERT_ID();
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Payment (booking_id, amount, payment_method)
    -> SELECT booking_id, total_amount, 'upi'
    -> FROM Booking
    -> WHERE booking_id = @new_booking_id;
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0
```

```
mysql> SET @new_payment_id = LAST_INSERT_ID();
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Escrow_Transaction
    -> (payment_id, status, platform_fee_percent, platform_fee_amount)
    -> SELECT payment_id, 'held', 5.00, ROUND(amount * 0.05, 2)
    -> FROM Payment
    -> WHERE payment_id = @new_payment_id;
Query OK, 1 row affected (0.00 sec)
Records: 1  Duplicates: 0  Warnings: 0

mysql>
mysql> SET SQL_SAFE_UPDATES = 0;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Equipment
    -> SET availability_status = 'booked'
    -> WHERE equipment_id = (
    ->     SELECT equipment_id
    ->     FROM Booking
    ->     WHERE booking_id = @new_booking_id
    -> );
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SET SQL_SAFE_UPDATES = 1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

- **STORED PROCEDURE**

```
mysql> DROP PROCEDURE IF EXISTS Calculate_Rental_Amount;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE Calculate_Rental_Amount(
    ->     IN p_equipment_id INT,
    ->     IN p_start_date DATE,
    ->     IN p_end_date DATE
    -> )
    -> BEGIN
    ->     DECLARE v_price DECIMAL(10,2);
    ->     DECLARE v_days INT;
    ->
    ->     SELECT price_per_day
    ->     INTO v_price
    ->     FROM Equipment
    ->     WHERE equipment_id = p_equipment_id;
    ->
    ->     SET v_days = DATEDIFF(p_end_date, p_start_date) + 1;
    ->
    ->     IF v_days <= 0 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'End date must be on or after start date';
    ->     END IF;
    ->
    ->     SELECT
    ->         p_equipment_id AS equipment_id,
    ->         p_start_date AS start_date,
    ->         p_end_date AS end_date,
    ->         v_days AS rental_days,
    ->         v_price AS price_per_day,
    ->         ROUND(v_price * v_days, 2) AS total_rental_amount;
    -> END$$
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> DELIMITER ;
```

```
mysql> CREATE PROCEDURE Release_Escrow(
    ->     IN p_booking_id INT
    -> )
    -> BEGIN
    ->     DECLARE v_payment_id INT;
    ->     DECLARE v_amount DECIMAL(10,2);
    ->     DECLARE v_fee_percent DECIMAL(5,2);
    ->     DECLARE v_fee_amount DECIMAL(10,2);
    ->
    ->     SELECT p.payment_id, p.amount
    ->     INTO v_payment_id, v_amount
    ->     FROM Payment p
    ->     WHERE p.booking_id = p_booking_id;
    ->
    ->     SELECT platform_fee_percent
    ->     INTO v_fee_percent
    ->     FROM Escrow_Transaction
    ->     WHERE payment_id = v_payment_id
    ->       AND status = 'held';
    ->
    ->     IF v_payment_id IS NULL THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'No payment found for this booking';
    ->     END IF;
    ->
    ->     IF v_fee_percent IS NULL THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'No held escrow found for this booking';
    ->     END IF;
    ->
    ->     SET v_fee_amount = ROUND(v_amount * v_fee_percent / 100, 2);
    ->
    ->     UPDATE Escrow_Transaction
    ->     SET status = 'released',
    ->         released_at = CURRENT_TIMESTAMP,
    ->         platform_fee_amount = v_fee_amount
    ->     WHERE payment_id = v_payment_id;
    ->
    ->     UPDATE Booking
    ->     SET status = 'returned'
    ->     WHERE booking_id = p_booking_id;
    ->
    ->     SELECT
    ->         p_booking_id AS booking_id,
    ->         v_amount AS total_amount,
    ->         v_fee_amount AS platform_fee,
    ->         ROUND(v_amount - v_fee_amount, 2) AS owner_payout;
    -> END$$
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> DELIMITER ;
```

- **TRIGGER**

```
mysql> DROP TRIGGER IF EXISTS trg_booking_status_update;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql>
mysql> DELIMITER $$
mysql>
mysql> CREATE TRIGGER trg_booking_status_update
    -> AFTER UPDATE ON Booking
    -> FOR EACH ROW
    -> BEGIN
    ->     IF OLD.status <> NEW.status THEN
    ->         INSERT INTO Booking_Status_Log
    ->         (booking_id, old_status, new_status)
    ->         VALUES
    ->         (NEW.booking_id, OLD.status, NEW.status);
    ->     END IF;
    -> END$$
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> DELIMITER ;
```

- **VIEW USAGE**

```
mysql> SELECT *
    -> FROM Available_Equipment_View
    -> ORDER BY price_per_day ASC;
+--------------+---------------+-----------------------------------------------------+-----------------+------------+--------------+-----------------+--------------------+
| equipment_id | title         | description                                         | category_name   | owner_name | price_per_day | condition_status | availability_status |
+--------------+---------------+-----------------------------------------------------+-----------------+------------+--------------+-----------------+--------------------+
|            4 | Camping Tent  | Four-person waterproof camping tent.                | Camping & Outdoor | Arjun Kumar |       500.00 | good            | available          |
|            3 | Gaming Console | Latest generation gaming console with two controllers. | Electronics     | Ananya Rao |      1000.00 | new             | available          |
+--------------+---------------+-----------------------------------------------------+-----------------+------------+--------------+-----------------+--------------------+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT *
    -> FROM Booking_Escrow_View
    -> ORDER BY booking_id DESC;
+------------+-----------------+-------------+------------+------------+--------------+----------------+------------+-----------+----------+---------------+---------------------+--------------------+
| booking_id | equipment       | renter_name | start_date | end_date   | total_amount | booking_status | payment_id | paid_amount | escrow_id | escrow_status | platform_fee_percent | platform_fee_amount |
+------------+-----------------+-------------+------------+------------+--------------+----------------+------------+-------------+-----------+---------------+---------------------+--------------------+
|          2 | Mountain Bike   | Meera Sharma | 2026-10-10 | 2026-10-12 |      2400.00 | confirmed      |          2 |     2400.00 |         2 | held          |                5.00 |             120.00 |
|          1 | Canon DSLR Camera | Meera Sharma | 2026-10-05 | 2026-10-07 |      3600.00 | confirmed      |          1 |     3600.00 |         1 | held          |                5.00 |             180.00 |
+------------+-----------------+-------------+------------+------------+--------------+----------------+------------+-------------+-----------+---------------+---------------------+--------------------+
2 rows in set (0.00 sec)

mysql>
```